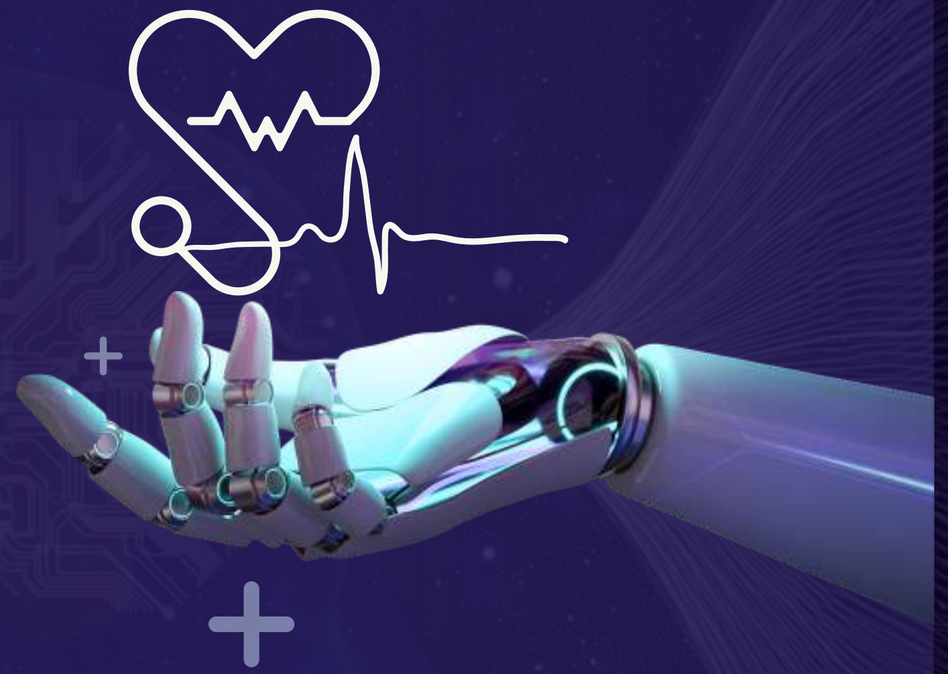
AI In Healthcare

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Introduction



- **Artificial Intelligence (AI)** encompasses machine learning and neural networks, technologies that enable computers to learn, reason, and solve problems like humans.



- The rapid evolution of ChatGPT from version 3.5 to 4 demonstrates the transformative potential of AI, improving comprehension and communication.

AI in Healthcare



- Artificial Intelligence in healthcare involves using algorithms and computer models to ***diagnose diseases***, ***predict treatment outcomes***, and ***enhance healthcare quality***.



- **AI Systems in Healthcare**
  - **Dendral:** Molecular structure inference system.
  - **CASNET:** Eye disease diagnosis.
  - **PIP:** Kidney disease diagnosis.
  - **ONCOCIN:** Cancer treatment protocol guide.



- **Global AI Healthcare Market Growth**
  - **2022 Valuation:** Estimated at $15.1 billion.
  - **2030 Projection:** Expected to exceed $187.95 billion.

Why AI in Healthcare?



- **Data:** The advent of electronic health records(EHRs) has provided access to a vast amount of accurate and relevant data.



- **Challenges:** The healthcare sector faces a wide range of challenges, creating numerous opportunities for the application and development of AI across various segments.



Challenges

- **Patient Scheduling and Registration:** Inefficient appointment scheduling and long wait times.
- **Check-in:** Manual data entry errors and delays in the check-in process.
- **Treatment administration:** Adherence issues, and limited access to doctors when problems arise.
- **Monitoring and follow-up:** Limited resources for continuous monitoring.
- **Medical records management:** Manual record-keeping increases the potential for data errors.
- **Patient education:** Limited resources for patient education.
- **Follow-up and aftercare:** Inadequate monitoring of post-treatment outcomes.

Digital Twin

A Digital Twin in healthcare is a virtual model of a patient or a medical process that uses real-time data and AI to simulate and predict health outcomes, enhancing personalized treatment and medical decision-making.

01

Data Collection

ncludes patient medical records, medical imaging, diagnostic tests, genetic information, and biometric data.

02

Digital Simulation

Develops an accurate, dynamic model of the patient that updates based on collected data.

03

Analysis and Optimization

AI algorithms analyze data to predict diseases, recommend treatments, and optimize health outcomes.

Project Overview

Utilizing artificial intelligence to create and tailor **personalized diet plans** for individual health optimization



01

- **Customized Diets:** Tailors nutrition specifically to individual health needs and preferences.

02

- **Interactive Consultation:** Provides expert advice and answers to dietary questions, enhancing user engagement and understanding.

03

- **Optimal Health Management:** Boosts overall wellness and prevents health issues through precision nutrition.



Digital Twin For Diet Project

This project employs **Digital Twin** technology to provide personalized diet plans tailored to the unique physiological conditions of each individual. This technology simulates real body responses, allowing for the optimization of dietary recommendations.

01 Data Collection

02 Digital Modeling

03 Analysis and Simulation

04 Nutritional Recommendations

05 Monitoring and Optimization

Digital Twin in Diet Optimization



Personalization

Offers diet recommendations tailored to individual physiology.



Prediction and Prevention

Prevents issues like obesity or high blood sugar before they arise



Enhanced Diet Adherence

Provides motivating, personalized dietary feedback



Immediate Feedback

Offers instant feedback based on changes in biometric data

THANK YOU
FOR YOUR ATTENTION

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